# Bihar Engineering University, Patna

## B.Tech 3rd Semester Examination, 2024

*Course: B.Tech* | *Subject: Surveying & Geomatics* | *Time: 03 Hours*
*Code: 101308* | *Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

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***Q.1 Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In plane table surveying, the method used to plot the position of an object directly on the drawing sheet is called
(i) Radiation (iii) Intersection
(ii) Traversing (iv) Leveling

(b) Which of the following is an error in levelling?
(i) Parallax (iii) Collimation error
(ii) Instrumental drift (iv) Refraction error

(c) The process of extending the base line in triangulation is done by
(i) Plane table (iii) Theodolite
(ii) Satellite station (iv) Cross staff

(d) The main function of a transition curve is to
(i) Reduce centrifugal force gradually (iii) Increase friction
(ii) Divide vertical curves (iv) Enhance contour accuracy

(e) Which of the following is a modern surveying equipment?
(i) Cross staff (iii) Plane table
(ii) Distomat (iv) Compass

(f) What is the primary function of a Total Station?
(i) Image processing (iii) Photogrammetry
(ii) Simultaneous measurement of angles and distances (iv) Only measuring elevation

(g) GPS surveying is affected by
(i) Cloud cover (iii) Solar radiation
(ii) Ionospheric and tropospheric delays (iv) Reflection from buildings only

(h) The principle of photogrammetry is based on
(i) Levelling (iii) Remote sensing
(ii) Perspective geometry (iv) Reflection

(i) Relief displacement in aerial photography depends on
(i) Altitude of the flight only (iii) Focal length of the camera
(ii) Both object height and camera height (iv) Weather conditions

(j) In remote sensing, which part of the electromagnetic spectrum is used for thermal imaging?
(i) Ultraviolet (iii) Visible
(ii) Microwave (iv) Infrared



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**Q.2** (a) Describe the methods of levelling and illustrate differential and reciprocal levelling with neat diagrams. [7]

(b) The following staff readings were taken with a level and a 4 m leveling staff. The instrument was shifted after the 4th and 7th readings. The readings are as follows in order: [7]
2.345, 1.955, 1.450, 2.875, 2.225, 1.675, 1.250, 2.300
The first reading was taken on a benchmark with a reduced level (RL) of 100.000 m.
Using the height of instrument method, prepare a level book and calculate the reduced levels of all points.

**Q.3** (a) Explain the concept of contouring. Describe the characteristics of contour lines and their uses. [7]

(b) The following spot levels were taken at 20 m intervals along a section line: [7]
145.2, 146.8, 147.5, 148.0, 147.0, 146.2
Calculate:
(i) Area enclosed between these points using the trapezoidal rule.
(ii)Volume of excavation if the depth of cutting is uniform at 2 m.

**Q.4** (a) Define the elements of a simple circular curve and explain the procedure to set it out in the field. [7]

(b) Two straight lines intersect at a deflection angle of 60°. A simple circular curve of radius 300 m is to be introduced to connect the two tangents. [7]
Calculate the following:
(i) Tangent length
(ii) Length of the curve
(iii) Long chord
(iv) Apex distance

**Q.5** (a) Explain the working principle of Electronic Distance Measurement (EDM) and describe the types of EDM instruments. [7]

(b) An EDM instrument measured a sloping distance of 650.000 m between two stations. The vertical angle from the instrument to the target was found to be +3°10′. [7]
Calculate the horizontal distance between the two stations and the difference in elevation.

**Q.6** (a) Illustrate the field procedure for conducting a Total Station survey. [7]

(b) Explain the principles of surveying and differentiate between plane and geodetic surveying with examples. [7]

**Q.7** (a) Describe the process of measuring horizontal and vertical angles using a theodolite. [7]

(b) In a closed traverse, the following observed angles were recorded using a theodolite: [7]
A = 92°30′, B = 88°45′, C = 96°00′, D = 82°30′.
(i) Find the included angles at each station
(ii) Check the angular closure
(iii) Apply correction, if any, using equal distribution method

**Q.8** (a) Explain the interaction of electromagnetic radiation with the atmosphere and earth surface in the context of remote sensing. [7]

(b) Discuss the procedure for aerial triangulation and its use in map preparation [7]

**Q.9** Write short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Reciprocal Levelling and its application in surveying across obstacles like rivers.
(b) Errors in Levelling and how they are minimized in fieldwork.
(c) Intervisibility in triangulation and the factors affecting it.
(d) Stereoscopy and its role in aerial photogrammetry.



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